Mission Sophie: The Stolen Powers

A birthday party game with 3D-printed NFC pendants



✓ Tested for real: the party was a hit!

For 7 to 10 kids aged 9-10, about 1.5 hours. Inspired by the *Keeper of the Lost Cities* series: the Neverseen have stolen the girls' talents. Each child gets a **3D-printed pendant with an NFC tag hidden inside**. Placed on an Android phone, "Sophie's Transmitter", it awakens her power and starts a challenge: a code to find, a cipher wheel, a shout to let out, a Neverseen trap… all the way to Sophie's Heart.

What you get (free)

- **18 drop-shaped pendants** with talent symbols, 2 colours, for NFC key-fob tags
- **2 fake badges** (the trap) and **Sophie's Heart** (the final star)
- An **elvish cipher wheel** with snap-fit discs (letters, or letters + digits)
- Parametric **OpenSCAD** files and all the **STLs**
- The **Android app** (offline, voices, music, video), the **game guide** (PDF) and printable **cards**

The printed parts

Pendants

- Size: 33 × 69 × 4.2 mm, loop with a 4.4 mm hole for a cord or a ring.
- Made for **drop-shaped NFC key-fob tags**, 27 × 54 mm, 1.1 mm thick (any NTAG213/215: the app only reads the tag ID). Measure yours and adjust `TAG_L`, `TAG_R_BIG`, `TAG_T` in the .scad file.
- **2 colours**: `<talent>-base.stl` (colour 1) and `<talent>-symbol.stl` (colour 2), loaded together: they overlap exactly.
- **Front face on the bed**, no supports: the symbol is already mirrored so it reads the right way.
- Single-colour printer: print the base only, the symbol shows engraved (and can be painted).
- Slide the tag into the pocket at the back, then glue or press-fit `couvercle.stl` (the lid).

Sophie's Heart and fake badges

The Heart (87 mm, no loop, with a border) is the final badge. The fake badges mimic a real pendant with a distorted symbol: that's the trap.





Elvish cipher wheel

- Large disc Ø 116 mm, small disc Ø 84 mm; inlaid letters in a 2nd colour (`*_lettres.stl`).
- The discs turn on a snap-fit axle. Print `roue-test_axe.stl` first: if the snap is too stiff, lower `LEVRE` (0.35 mm tested).
- `chiffres/` version: 36 slots (A–Z, Ø, 1–9) for codes like K6.



> **Suggested settings:** PLA, 0.2 mm layers, 2-3 walls, 15-20 % infill, no supports.

How to play

1. Download the kit and the APK from GitHub (link below), install the app on an Android phone with NFC.
2. Set the **name**, **age** and **hiding places** in the app (Admin › Réglages › Personnaliser).
3. Pair each tag with its pendant (Admin › Liaison), print the cards, set up the challenges with the guide.
4. On the day, mirror the phone to the TV and let Sophie lead the kids!

> The app and the guide are in **French**; all the texts live in a single file and can be translated. The 3D files work for any game.

Variant: round medallions

In the .scad file, `SHAPE = "rond"` gives round medallions with a larger symbol.



Printable cards (front)



EVERYTHING IS HERE

github.com/tigris25/mission-sophie-kit

3D files, app, guide, cards and sources

License

Models, texts and documents: **CC BY-NC 4.0** (share and remix with credit, no commercial use). Code: MIT.

Unofficial, free fan project, not affiliated with Shannon Messenger or her publishers. Sophie's voices, portrait and video are AI-generated. Music: "Fantascape", Eric Matyas, www.soundimage.org. Symbols based on a child's drawings.